

Project Report: Hospital Network Design

Project Title:

Hospital Network Simulation – Designed and Simulated using Cisco Packet Tracer

Project Objective:

To design and simulate a secure, scalable, and fully functional enterprise network for a hospital using Cisco Packet Tracer. The design follows real-world enterprise networking practices and healthcare IT requirements, focusing on segmentation, high availability, centralized services, and security.

Phase 1 – Identify Needs and Requirements

Business Understanding

The hospital aims to deliver uninterrupted, secure, and reliable healthcare services. Its IT infrastructure must support:

- **Vision & Mission:** Reliable and secure healthcare delivery, seamless information flow, and patient data protection.
- **Goals:**
 - Provide highly available and secure access to Electronic Medical Records (EMR).
 - Enable efficient communication across departments.
 - Support scalability for future growth.

Departments

- Administration
- Medical Staff
- Pharmacy
- Laboratory
- Emergency Unit

User Base & Functions

- **Administration:** Staff PCs for patient registration, billing, and reporting.
- **Medical Staff:** Workstations, tablets, and laptops accessing EMR, medical imaging, and patient records.
- **Pharmacy:** Inventory management systems and prescription access.
- **Laboratory:** Access to diagnostic systems, result processing, and reporting tools.
- **Emergency Unit:** High-priority, real-time access to patient data and communications.

Existing Network Study

No enterprise-grade existing infrastructure. The project introduces a new, secure, and robust hospital-wide enterprise network.

Site Layout & Datacenter

- A centralized **Datacenter/Server Room** hosts key internal applications (Web, Mail, EMR).
- A **DMZ zone** isolates public-facing and sensitive internal services.
- Departmental floors connected via **Access Switches and Wi-Fi APs**.

Physical Layout (Simplified)

- Core layer: Dual Core Switches with redundancy.
 - Distribution layer: VLAN segmentation per department.
 - Access layer: Five Access Switches + Wi-Fi Access Points.
 - Firewall layer: Cisco ASA between internal network and Internet.
 - Edge router connected to simulated ISP.
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Phase 2 – Sketch the Network Architecture

User Devices (Access Computing Devices)

- PCs, laptops, tablets per department.
- Wi-Fi Access Points for wireless access.

User Applications & Server Applications

- **User Applications:** EMR system, email, web portal, lab diagnostic tools, billing & inventory apps.
- **Server Applications:** Web server, Mail server, EMR server, DNS, DHCP.

Access Devices

- **Access Switches:** 5 (one per department).
- **Wireless Access Points:** One per department.
- **Core Switches:** 2 (HSRP enabled).
- **Edge Router:** 1 (connected to ISP).
- **Cisco ASA Firewall:** Security gateway.

Logical Networks (VLANs)

- VLAN 10 – Administration (192.168.10.0/24)

- VLAN 20 – Medical Staff (192.168.20.0/24)
- VLAN 30 – Pharmacy (192.168.30.0/24)
- VLAN 40 – Laboratory (192.168.40.0/24)
- VLAN 50 – Emergency (192.168.50.0/24)
- VLAN 60 – DMZ Servers (192.168.60.0/24, on ASA)
- VLAN 99 – Native VLAN for management/trunks

Core & Distribution Channels

- **Core Layer:** Dual Core Switches with HSRP for redundancy.
- **Routing Protocol:** OSPF on Core Switches, Edge Router, and ASA Firewall for dynamic routing and fast convergence.
- **Load Balancing/Redundancy:** HSRP with preemption and priority ensures seamless failover.

External Connections

- ISP/Internet connection through the Edge Router.
- ASA Firewall provides NAT for internal → Internet communication and static NAT for DMZ services.

Security Components

- Cisco ASA Firewall with:
 - Dynamic & Static NAT.
 - ACLs restricting inbound traffic to allowed services (HTTP/HTTPS/SMTP).
 - ICMP inspection for ping testing.
- VLAN segmentation to isolate departmental traffic.

Quality of Services (QoS)

- Emergency Unit VLAN prioritized for real-time and critical applications.
- Segmentation ensures fair bandwidth distribution.

Server Applications & Hardware

- Internal servers in DMZ (Web, Mail, EMR, DNS).
- DHCP server role configured on Core Switch-1 (with backup on Core-2).

Phase 3 – Design Considerations

IP Network Design & Addressing Scheme

VLAN	Department	Subnet	Gateway (HSRP VIP)
10	Administration	192.168.10.0/24	192.168.10.1
20	Medical Staff	192.168.20.0/24	192.168.20.1
30	Pharmacy	192.168.30.0/24	192.168.30.1
40	Laboratory	192.168.40.0/24	192.168.40.1
50	Emergency	192.168.50.0/24	192.168.50.1
60	DMZ Servers	192.168.60.0/24	192.168.60.1 (ASA)
99	Native VLAN	N/A	N/A

- DHCP Excluded: .1–.10 reserved (with .1 as HSRP VIP).
- Client IPs allocated dynamically from DHCP pools.

Configuration Scheme

1. Client Devices

- DHCP assigned IPs, DNS, and default gateway.

2. Switches (Core & Access)

- VLANs configured and trunking enabled.
- HSRP configured on Core Switches for redundancy.
- Inter-VLAN routing on Core Switches.

3. Routers & Security Devices

- Edge Router: OSPF and ISP simulation.
- ASA Firewall:
 - Inside, Outside, and DMZ interfaces configured.
 - NAT (static and dynamic).
 - ACLs for controlled access.

4. Servers

- Web, Mail, EMR, DNS in DMZ subnet.
- Accessible internally and externally via static NAT.

5. External Connections

- Internet simulated through router.
- ASA provides secure boundary.

6. Backup & Security Measures

- Redundant Core Switches with HSRP.
- OSPF dynamic routing for fault tolerance.
- DMZ isolation for public-facing servers.
- ACLs and inspection policies to secure data flows.

Conclusion

The **Hospital Network Simulation** meets the organizational requirements of hospital by delivering a **secure, redundant, and scalable** infrastructure. The integration of **HSRP, OSPF, VLAN segmentation, ASA Firewall with NAT/ACLs, and DMZ-based services** ensures:

- High availability and fault tolerance.
- Segregation of traffic for efficiency and security.
- Centralized and secure access to critical healthcare applications.
- Real-world enterprise-grade practices applied in a simulated environment.

This network provides a **future-proof foundation** for hospital IT operations, ensuring reliable connectivity and safeguarding patient data.